THE POINT CONDITIONS BETA, GAMMA, DELTA, EPSILON AND THETA PULL BACK ALONG REPRESENTABLE MORPHISMS

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Editorial note. The theta case is a one-line citation of a Categories lemma (cross-chapter, named in the prerequisites); the base-change boundedness input is not carded and is declared external.

Allowed standard prerequisites (editorial).

- UR-finiteness characterization. lemma-UR-finite-above-x (carded L26700). Condition (2) expands (gamma) into the covering-family form.
- U-universally-bounded characterization. lemma-U-universally-bounded (carded L26701). Closes the delta case.
- Base change of universal boundedness. lemma-base-change-universally-bounded (same chapter, NOT carded, declared external). Gives bounded fibres of V x_Y X -> X in the delta case.
- Very-reasonable characterization. lemma-characterize-very-reasonable (carded L26702). Expands (epsilon) into the covering-family form.
- Base-change transfer for representable transformations. Spaces, lemma base-change-representable-transformations-property (cross-chapter, named). Used throughout the epsilon case.
- Representability over representable. Categories, lemma representable-over-representable (cross-chapter, named). The theta case.

Source reference. Original reference 03KC: focused-seeds/geometry/S9-decent-representable-properties/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JT: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 847–847. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 732–732. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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Source reference. Original lemma 03JN: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 327–327. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JX: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 952–952. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03IH: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 896–896. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02WL: focused-seeds/geometry/S9-decent-representable-properties/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{P}$ be one of the properties (β), (γ), (δ), (ϵ), or (θ) of algebraic spaces listed in Lemma 03JX. Let X, Y be algebraic spaces over S. Let $X \to Y$ be a representable morphism. If Y has property $\mathcal{P}$, so does X.*

Proof. Assume $f : X \to Y$ is a representable morphism of algebraic spaces, and assume that Y has $\mathcal{P}$. Let $x \in |X|$, and set $y = f(x) \in |Y|$.

The case $\mathcal{P} = (\beta)$. Condition (β) for Y means there exists a monomorphism $\mathrm{Spec}(k) \to Y$ representing y. The fibre product $X_y = \mathrm{Spec}(k) \times_Y X$ is a scheme, and x corresponds to a point of X_y, i.e., to a monomorphism $\mathrm{Spec}(k') \to X_y$. As $X_y \to X$ is a monomorphism also we see that x is represented by the monomorphism $\mathrm{Spec}(k') \to X_y \to X$. In other words (β) holds for X.

The case $\mathcal{P} = (\gamma)$. Since $(\gamma) \Rightarrow (\beta)$ we have seen in the preceding paragraph that y and x can be represented by monomorphisms as in the following diagram

$$\begin{array}{ccc} \mathrm{Spec}(k') & \xrightarrow{x} & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \xrightarrow{y} & Y \end{array}$$

Also, by definition of property (γ) via Lemma 03JV (2) there exist schemes V_i and étale morphisms $V_i \to Y$ such that $\coprod V_i \to Y$ is surjective and for each i, setting $R_i = V_i \times_Y V_i$ the fibres of both

$$|V_i| \longrightarrow |Y| \quad \text{and} \quad |R_i| \longrightarrow |Y|$$

over y are finite. This means that the schemes $(V_i)_y$ and $(R_i)_y$ are finite schemes over $y = \mathrm{Spec}(k)$. As $X \to Y$ is representable, the fibre products $U_i = V_i \times_Y X$ are

schemes. The morphisms $U_i \to X$ are étale, and $\coprod U_i \to X$ is surjective. Finally, for each i we have

$$(U_i)_x = (V_i \times_Y X)_x = (V_i)_y \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k')$$

and

$$(U_i \times_X U_i)_x = ((V_i \times_Y X) \times_X (V_i \times_Y X))_x = (R_i)_y \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k')$$

hence these are finite over k' as base changes of the finite schemes $(V_i)_y$ and $(R_i)_y$. This implies that (γ) holds for X, again via the second condition of Lemma 03JV.

The case $\mathcal{P} = (\delta)$. Let $V \to Y$ be an étale morphism with V an affine scheme. Since Y has property (δ) this morphism has universally bounded fibres. By Lemma 03JN the base change $V \times_Y X \to X$ also has universally bounded fibres. Hence the first part of Lemma 03JT applies and we see that Y also has property (δ).

The case $\mathcal{P} = (\epsilon)$. We will repeatedly use Spaces, Lemma 02WL. Let $V_i \to Y$ be as in Lemma 03IH (2). Set $U_i = X \times_Y V_i$. The morphisms $U_i \to X$ are étale, and $\coprod U_i \to X$ is surjective. Because $U_i \times_X U_i = X \times_Y (V_i \times_Y V_i)$ we see that the projections $U_i \times_Y U_i \to U_i$ are base changes of the projections $V_i \times_Y V_i \to V_i$, and so quasi-compact as well. Hence X satisfies Lemma 03IH (2).

The case $\mathcal{P} = (\theta)$. In this case the result is Categories, Lemma 03KC. □